

[Gamil Mamdouh El Gamil Asham]

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PROFILE

STEM student at Assiut STEM School (expected graduation 2027) with three years of experience in the Digital Egypt Cubs Initiative and a portfolio of independent embedded-systems, automation, and web/software projects. Experienced in designing, building, and testing functional hardware and software prototypes — from ESP32-based sensor control systems to serverless web platforms — with an emphasis on measurable performance, research, and iterative debugging.

EDUCATION

Assiut STEM School

GPA: 3.5 (Grades 10–11) | Grade 12 GPA: [To be added]

TECHNICAL SKILLS

Languages: Python, C/C++, MicroPython, HTML/CSS/JavaScript

Platforms & Tools: ESP32, Raspberry Pi Pico, Arduino, Wokwi, CoppeliaSim, Tinkercad, Firebase, Git/GitHub

SELECTED TECHNICAL PROJECTS — ASSIUT STEM SCHOOL (TEAM)

Multi-Parameter Feedback Control for Automated Greenhouse Systems

Grade 11, 2025/2026

- Co-developed an ESP32-based greenhouse automation system monitoring soil moisture, air humidity, and irrigation water TDS/salinity across two crop profiles (Basil, Okra); contributed to research, prototype construction, and documentation.
- Implemented rule-based IF-THEN control logic with 10-second sampling and 2-minute hysteresis; system operated autonomously for 6+ hours per session and responded to threshold breaches within ≤ 5 seconds.
- Testing showed humidity reduced from 84.3% to 73.2% in ~ 4 minutes, TDS from 1748 to 1487 ppm in ~ 15 minutes, and soil moisture recovered from 34.8% to 63.5% in ~ 16 minutes, at 8.5 W peak actuator power (~ 1.10 Wh/session); all 12 design requirements were met.

Rollmetric — Digital Measuring Wheel & Inclinometer

Grade 11, 2025/2026

- Built a combined distance/angle measurement device using an ESP32, magnetic rotary encoder wheel, dual Hall-effect sensors, and an MPU6050 accelerometer; reused electronic and mechanical materials to reduce e-waste.
- Diagnosed a magnet-spacing design flaw and corrected it ($5 \rightarrow 10$ magnets, spacing 0.999 ± 0.0004 cm), reducing distance measurement error from 33.3% to 0.2%; final angle error 4.2%.

Solar-Powered Pumped Hydro Storage (PHS) System

Grade 10, 2024/2025

- Designed a solar-charged closed-loop energy storage system (18V/10W solar panel, 12V DC pump, water turbine generator, lithium batteries), modeled in Tinkercad and built largely from recycled materials.
- Achieved 87.1 J of stored energy in 5 minutes and reached the 40 J design target in ~ 2.5 minutes, within a footprint under 0.5 m^3 .

Hydraulic Multi-Functional Bridge

Grade 10, 2024/2025

- Contributed to bridge design, research, and load/force calculations for a multi-use concept combining vehicle, pedestrian, and cyclist traffic with a working hydraulically actuated movable span for boat passage.
- Helped construct the physical prototype using palm frond/jreed as a construction material; the movable section was functional.

DIGITAL EGYPT CUBS INITIATIVE (3 YEARS — INDIVIDUAL PROJECTS)

Completed 5 individual embedded-systems and software projects, each carried out independently from design through debugging, as part of a nanodegree-style program managed by Udacity and sponsored by Egypt's Ministry of Communications and Information Technology (MCIT):

- Smart Delivery System — ESP32/MicroPython access-controlled delivery station paired with an autonomous CoppeliaSim/Python delivery robot (Pioneer3DX) featuring obstacle avoidance and destination detection.
- Smart Parking System — ESP32 non-blocking control logic for an automatic entrance barrier using ultrasonic distance sensing and consecutive-reading validation.
- Embedded Voltmeter — Raspberry Pi Pico ADC-based voltmeter with signal averaging, button debouncing, and a seven-segment display.
- Digital Device-Control Interface — MicroPython/Wokwi passcode-controlled interface applying digital logic (AND/OR/NOT gates, multiplexing, binary decoding).

- Adventure Game — Python text-based adventure game with branching outcomes, input validation, and game-state tracking.

INDEPENDENT / FREELANCE PROJECTS

- Online Exam / Quiz System — Serverless Firebase-based exam platform supporting 5 question types, per-question and master timers, cheating-detection monitoring, and a live admin dashboard with manual score override.
- St. Mark's Church Hymn Ministry Management System — Single-file, Firebase-backed Arabic (RTL) system managing students, attendance, exams/grades, and leaderboards across 8 hymn skill levels.
- Thyatira Team Entertainment Streaming Platform — Firebase-authenticated theater-content streaming platform with an access-code admin dashboard and anti-piracy deterrence mechanisms.

LEADERSHIP & SCHOOL ACTIVITIES

- Co-organized and delivered Computer Science sessions/presentations for peers at Assiut STEM School.
- Contributed to organizing school Ramadan activities and participated in multiple school Fun Days.
- Participated in the school football league.

CERTIFICATIONS (37 TOTAL — SELECTED; FULL LIST IN PORTFOLIO)

- Digital Egypt Cubs Initiative, Level Three: Embedded Systems and Robotics — Udacity / Egypt MCIT (Oct 2025)
- Digital Egypt Cubs Initiative, Level Two — Udacity / Egypt MCIT (Oct 2024)
- Introduction to Modern AI — Cisco Networking Academy (Aug 2026)
- Microprocessors and Microcontrollers, Professional Certificate Program — Academy Europe Open University (Dec 2025)
- Python Developer; SQL Intermediate; Introduction to SQL — Sololearn (2025)
- Cloud Computing, AI Basics, Data Management & Analysis, Python Programming Basics, Information Representation — Huawei ICT Academy CRA Training Program (Dec 2025)

ENGLISH PROFICIENCY

Duolingo English Test — Taken; Score: [To be added]